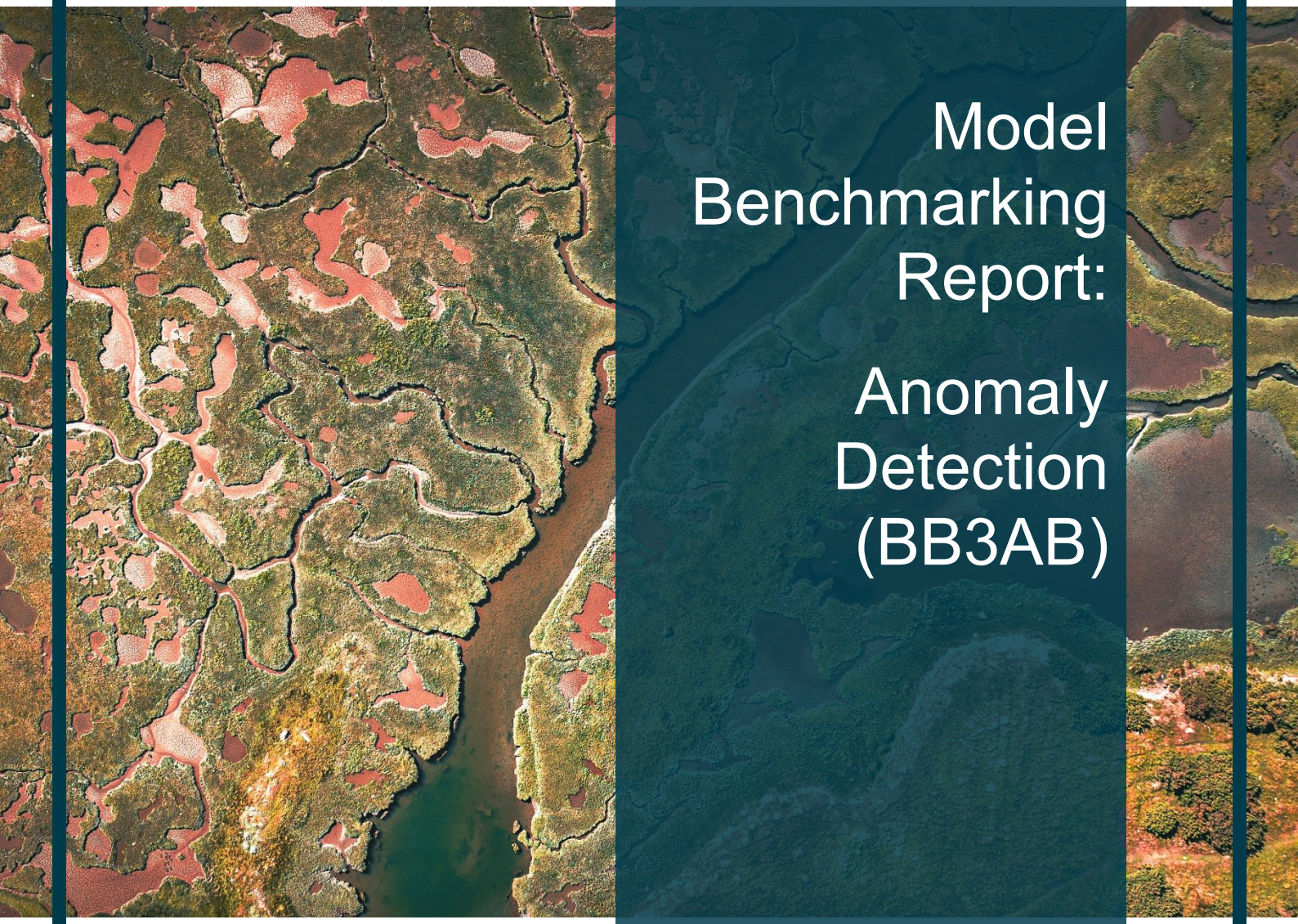
River Deep Mountain AI



Model Benchmarking Report:

Anomaly Detection (BB3AB)

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March 2026

Model Benchmarking Report - Anomaly Detection (BB3AB)

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Glossary

DO	Dissolved Oxygen
E2E	End-to-end
IoT	Internet of Things
LOESS	Locally Estimated Scatterplot Smoothing
ML	Machine Learning
NTU	Nephelometric Turbidity Unit
SQL	Structured Query Language
SVM	Support Vector Machine

Summary

Model Outputs	**Google BigQuery** (Existing Industry Model)	RDMAI Model
Model Approach	BigQuery uses the ARIMA_PLUS_XREG, K-means and PCA machine learning (ML) models for anomaly detection. The ARIMA_PLUS_XREG model is a forecast model. Anomaly detection is therefore predictive, by comparing the predicted main timeseries with observations and flagging anomalies with respect to their placement within or outside the expected predictive range. The ARIMA_PLUS_XREG is multivariate in that it uses other timeseries 'external regressors' to predict a single timeseries. Anomaly detection in ARIMA_PLUS_XREG is different to that in the RDMAI model in that you need to stipulate a main timeseries to forecast in ARIMA_PLUS_XREG, whereas the RDMAI model identifies anomalies across all the timeseries simultaneously. In this benchmarking, two ARIMA_PLUS_XREG models for each monitoring point were constructed: for predicting DO and ammonia, respectively. Anomalies for a monitoring point were then determined by combining model identified anomalies for ammonia and DO. The PCA and K-means models are more similar to the RDMAI model in that they highlight an anomaly in the combination of values in a row. However, they don't remove trend and seasonality of the time series before performing anomaly detection. In addition, since the variables in the monitoring data are not independent, the PCA and K-means models are not ideal models to apply.	The RDMAI model uses a ML approach, incorporating infilling missing data, separating data residuals from seasonal patterns and trends, and identifying anomalies in the residuals through the Isolation Forest unsupervised algorithm. The RDMAI model additionally performs clustering and filtering of anomalies, although the usefulness of this function remains limited at this stage, as the clusters are not linked to any type of event. This RDMAI anomaly detection model overcomes several challenges associated with traditional approaches by providing a framework that enables more precise estimation of long-term trends and recurring seasonal patterns.

Model Outputs	**Google BigQuery** (Existing Industry Model)	RDMAI Model
Applications	ARIMA_PLUS_XREG: Anomaly detection in timeseries data with exogenous factors. PCA and K-means: Anomaly detection in independent and identically distributed random variables. The models are generic and can be applied to a wide range of data, ranging from financial data to environmental data.	Anomaly detection and clustering in high-frequency water quality monitoring data, with a focus on the Section 82 parameters – dissolved oxygen, temperature, pH, turbidity, and ammonia, as well as conductivity.
Geographic Constraints	The model is generic and has no geographic constraints.	The model is generic and has no geographic constraints.
Temporal Resolution	The ARIMA_PLUS_XREG: model can be applied to any timeseries data. The PCA and K-means models can be applied to any independent and identically distributed data.	Timeseries water quality monitoring data with a frequency of 1 hour or less should be used, as the model averages data to hourly during pre-processing.
Data Inputs Required	Time series data or independent and identically distributed random variables.	Time series high-frequency water quality monitoring data for pH, dissolved oxygen (mg/L or %), turbidity (NTU), temperature (°C), and ammonia (mg/L).
Model Performance	For ARIMA_PLUS_XREG and PCA, clusters of anomalies broadly lined up with labelled clusters of anomalies. When comparing model identified anomalies with labelled data, Google ARIMA_PLUS_XREG and PCA identified between 14.2% and 62.3% of anomalous events. The model performances mentioned above should NOT be regarded as indicative of absolute model performance given	The RDMAI model clusters of anomalies broadly lined up with labelled clusters of anomalies. When comparing model identified anomalies with labelled data, the RDMAI model identified between 20.2% and 60.4% of anomalous events. The model performance mentioned above should NOT be regarded as indicative of absolute model performance given shortcomings in the performance evaluation approach, but rather as relative performances when comparing models.

Model Outputs	**Google BigQuery** (Existing Industry Model)	**RDMAI Model**
	shortcomings in the performance evaluation approach, but rather as relative performances when comparing models. It should be noted that domain experts labelled anomalies in each time series individually, which is a benchmarking limitation.	It should be noted that domain experts labelled anomalies in each time series individually, which is a benchmarking limitation.
Cost & Time	Learning to use BigQuery is a learning curve, particularly for those not used to SQL. Once familiar with the platform and the SQL commands for making tables, bringing in data, and running queries, the anomaly detection queries take seconds. BigQuery is part of Google's Cloud Services offering, which is a paid-for service. On the 'on-demand' payment plan, costs depend on usage and can quickly escalate. Pricing Information: • On-demand queries: $6.25 per TB processed Assuming a water company has 4000 sondes, anomaly detection every hour based on the previous years' worth of monitored data would cost about £42,073.	Installing and running the RDMAI Anomaly Detection Model requires a basic knowledge of Python and will take between 1 and 2 hours. Once the model is installed, running the anomaly detection model on around 2 months of data takes seconds. The RDMAI model is free to use and open access (https://github.com/Cognizant-RDMAI/Open-Anomaly-Detection-Model).

Model Outputs	**Google BigQuery** (Existing Industry Model)	RDMAI Model
	There are also costs for storage, but these are possibly not applicable as storage would be needed regardless of the model applied.	

Strengths	• The tool operates in the cloud, and there is no need for software installation or management.
	• Offers analysis of large volumes data in real time and near real time streamed from Internet of Things (IoT) devices, such as sondes.
	• Offers the flexibility of users creating and training their own models or using pre-built models.
	• It would be easier for most organisations already using Google Cloud Services to integrate the anomaly detection into their workflow for real-time or near real-time identification of anomalies.
	• Documentation for using the tool is comprehensive.
		• The RDMAI Anomaly Detection Model is open and free. The advantages of this include:
		  ○ Cost savings
		  ○ Easy customisation and improvement
		  ○ Readily embedded in water company data systems/platforms
		• The RDMAI Anomaly Detector is designed to detect anomalies in water quality monitoring data, primarily Section 82 data, whereas other industry standard options are more generic.
		• The functionality of the RDMAI Anomaly Detector extends beyond anomaly detection. It also filters anomalies to identify patterns and to group similar anomalies across multiple parameters using clustering and PCA. While the usefulness of this is currently limited as the clusters are not yet linked to events, the model could theoretically be improved to link clusters to events.

Model Outputs	**Google BigQuery** (Existing Industry Model)	**RDMAI Model**
		• The RDMAI anomaly detector scripts do produce some graph outputs that are more polished and easier for users to understand, whereas the outputs of the industry standard tools assessed need to be processed and graphed.
Limitations	• Google Cloud Services, of which BigQuery is a part, is a paid for service. If using an on-demand payment plan, costs can quickly escalate. • ARIMA_PLUS_XREG is a model often used for forecasting; therefore, anomaly detection is prediction based. In addition, as a forecasting model, the anomaly detection is in a single specified main timeseries, whereas the RDMAI model identifies anomalies in multiple timeseries simultaneously. • The BigQuery anomaly detection is generic, for any sort of data, and has not been designed to focus on water quality data. • The PCA and K-means models in BigQuery assume independent and identically distributed random variables and are therefore not ideal for application to water quality monitoring data.	• The RDMAI model would need to be modified to be able to be integrated into any sort of production environment because currently it is designed to work with historical data. • The RDMAI Anomaly Detector model offers an algorithm for detecting and classifying anomalies in high-frequency water quality monitoring data. However, it would need to be integrated and engineered for cloud-based IoT data-streaming and storage system for use by water companies for real-time analysis of Section 82 water quality monitoring data. Commercial options for cloud-based storage, such as Google or Microsoft, offer their own anomaly detection methods that are already integrated. • Unlike the industry standard tools, some basic knowledge of Python is needed to install and run the RDMAI anomaly detector.

Model Outputs	**Google BigQuery** (Existing Industry Model)	RDMAI Model
	• K-means in BigQuery requires the parameter, NUM_CLUSTERS', to be set, which can be related to pollution events, and an appropriate value for this parameter is not known beforehand. • The setting of the 'contamination' parameter in ML.Detect_Anomalies was set to 0.1, assuming that 10% of data in each dataset was anomalous. However, this parameter is uncertain as we do not have manual labelling across the entire labelled dataset. • Users would need to be proficient in SQL to use Google BigQuery.	

1. Benchmarking Methodology

1.1 Model Selection

1.1.1 RDMAI Anomaly Detection Model Overview

Water companies in England will be required to continuously monitor upstream and downstream of sewerage outfalls under Section 82 of the Environment Act of 2021. To meet this requirement, it is estimated that approximately 40,000–50,000 multi-parameter sondes will need to be deployed across England (upstream and downstream of the estimated 14,000 storm overflows and 9,000 sewage treatment works)[1]. These sondes will be recording data at a high frequency (every 15 minutes at high-risk times and every hour at other times), generating millions of data points annually. Detecting anomalies in the monitoring data associated with pollution events will be challenging due to:

- The sheer volume of data: this precludes the viability of manual human analysis of the data.
- Isolating anomalies in the monitoring data due to pollution events from natural fluctuations or trends or sensor drift.

The aim of the RDMAI Anomaly Detection Model is to provide an open access and free tool for rapidly and automatically detecting anomalies in high-frequency water quality monitoring data.

The Anomaly Detection model presents an end-to-end (E2E) framework designed to ingest, process, and analyse multi-parameter water quality data for UK water utilities, specifically focusing on the Section 82 parameters: pH, dissolved oxygen, turbidity, temperature and ammonia. The model can also consider conductivity.

The RDMAI Anomaly Detection Model uses a machine learning (ML) approach and builds a robust, end to end pipeline designed to isolate seasonal and trend signals, with the following broad steps:

- **Ingestion of multivariate water quality data**: Computes hourly means for each parameter, ensuring a common hourly timestamp index across all parameters; checks for missing data.
- **Identifying data range and handling missing data**: short gaps of missing data (< 24 h) are filled using linear interpolation, with the user notified if gaps exceeding 24 h are identified.

1. Adler & Allan (n.d.) The challenges of delivering Section 82 of the Environment Act. Adler & Allan. Available at: https://www.adlerandallan.co.uk/knowledge/insight/challenges-delivering-section-82-environment-act (Accessed: 4 March 2026).

- **Identifying trends and seasonality**: the seasonal and trend components in the time series of each parameter are isolated and removed. Seasonal cycles are detected using MSTL [Multi-Seasonal Trend decomposition using locally estimated scatterplot smoothing (LOESS)]. Long-term trends are identified using harmonic regression and Butterworth filtering.

- **Identifying residuals**: residuals are computed by removing seasonality and long-term trends from the observed data (*[raw signal - (seasonal + trend)]*) and then plotted. Residuals are computed for each parameter individually. Here, the 'raw signal' is the original monitored water quality timeseries data after hourly averaging and missing data infilling.

- **Anomaly detection**: The Isolation Forest unsupervised anomaly detection algorithm is fitted on all residual columns (residuals for each water quality parameter) simultaneously to detect multivariate anomalies.

- **Anomaly detection plotting**: The time series residuals of each parameter are plotted with detected joint (multivariate) anomalies overlayed on each plot shown as red points.

- **Anomaly filtering**: Clustering and Principal Component Analysis are applied to the multivariate data points flagged as anomalous by the Isolation Forest. This helps to identify patterns and to group similar anomalies across multiple parameters. The derived information can help identify similar or recurring events.

Due to project timescale constraints, the model has not gone through extensive validation. However, model validation showed that the RDMAI Anomaly Detection Model can detect short-term, high and low magnitude deviations from a baseline, but is not designed to detect long-term, low magnitude events or sensor drift. In addition, the anomaly filtering (clustering) provides limited information as the clusters are not linked to a type of event.

More information on the RDMAI Anomaly Detection Model is available on the RDMAI GitHub page (https://github.com/Cognizant-RDMAI/Open-Anomaly-Detection-Model).

1.1.2 Overview of Existing Industry Model(s)/Tool(s)

This benchmarking considered various anomaly detection tools. These included commercially available anomaly detection tools, such as Google BigQuery, the Azure anomaly detector (which was determined to be no longer supported) and the Azure Data Explorer. The benchmarking also briefly investigated anomaly detection integrated within the water quality monitoring systems of various water companies, including Anglian Water, Yorkshire Water, United Utilities, Thames Water, Wessex Water. The water companies contract water quality modelling and data interpretation out to companies such as WATR (https://www.watr.tech/), Meteor Communications (https://meteorcommunications.co.uk), Aperio (https://aperio.ai/aperio-datawise/) and FYLD (https://fyld.ai/). As such, the anomaly detection

used is proprietary, and the benchmarking did not have sufficient time to communicate with water companies to try obtain access to the models. The benchmarking also briefly investigated anomaly detection methods for water quality data published in the scientific literature.

Some publications, such as Liu et al. (2020) [1] and Leigh et al. (2019) [2] do not provide open access to the models. One publication, Rigueira et al. (2025) [3], does provide access to the model on GitHub (https://github.com/xrigueira/w-fda with DOI). Given constraints to the benchmarking timeline, the benchmarking considered the two commercially available anomaly detection options, Google BigQuery and Microsoft Azure Data Explorer. While the Azure Data Explorer looks promising, it was discovered that only a paid-for subscription allows access to multivariate anomaly detection. Therefore, this benchmarking concentrated on anomaly detection in Google BigQuery.

Google BigQuery

Description

BigQuery is part of the Google Cloud Platform. It is a cloud-based tool for analysis of large datasets. Since everything is on the cloud, the platform allows users to perform complex analysis on large quantities of data without having to install any software or hardware.

Data queries on BigQuery are through Structured Query Language (SQL), a relational database language used to query and manipulate records in database tables.

BigQuery can handle petabyte-scale datasets, executing queries in seconds or minutes by distributing the workload across multiple servers.

BigQuery offers Integrated Machine Learning, where users can create and train machine learning models directly. It also offers pre-built Models, such as ARIMA_PLUS for time series data or K-means clustering for unsupervised anomaly detection.

Costing

Google BigQuery uses a combination of on-demand or flat-rate pricing models, along with separate fees for storage, streaming inserts, and data export. There is a useful website (https://glowcalculator.com/bigquery-cost-calculator/) that helps you estimate costs. BigQuery can be tried for free with $300 in credit over 90 days.

1. Liu et al. (2020). An integrated data-driven framework for surface water quality anomaly detection and early warning. Journal of Cleaner Production, Vol 251.

2. Leigh et al. (2019). A framework for automated anomaly detection in high frequency water-quality data from in situ sensors. Science of the Total Environment, Vol 664.

3. Riqueira et al. (2025). Multivariate functional data analysis and machine learning methods for anomaly detection in water quality sensor data. Environmental Modelling and Software. Vol 190

Anomaly detection used

Anomaly detection in BigQuery is performed using ML.DETECT_ANOMALIES. There are various models you can use within ML.DETECT_ANOMALIES depending on the type of data you want to analyse. ML.DETECT_ANOMALIES forms part of BigQuery's machine learning capabilities and enables users to directly conduct anomaly detection using SQL queries. Since ML.Detect_Anomalies is integrated into the BigQuery environment, users can leverage the powerful data processing capabilities of BigQuery in anomaly detection. Users can easily create and execute machine learning models using SQL, making it accessible for those familiar with database queries.

ARIMA_PLUS_XREG is a multivariate time series forecasting model used in Google BigQuery. When setting up the model, one needs to specify the timeseries variable to forecast, and then other relevant variables (in this case, the other water quality parameters in the monitoring dataset) are set to be predictor time series variables. ARIMA_PLUS_XREG can detect seasonal patterns and trends. As a forecasting model, ARIMA_PLUS_XREG is different from the RDMAI Anomaly Detection model, as you need to specify a main time series for forecasting. Detecting anomalies in ARIMA_PLUS_XREG is prediction based: the model learns the patterns of the time series to predict expected values. It then compares the fitted curve to the observed data to the model's predicted confidence interval. Anomalies are identified as points that fall significantly outside of the expected range. In other words, the model attempts to predict what the data timeseries 'should be' and then compares this to the observed data to identify any anomalies. The anomaly detection in ARIMA_PLUS_XREG is different to that in the RDMAI model as the ARIMA_PLUS_XREG identifies anomalies in a specified single timeseries, whereas the RDMAI model identifies anomalies across multiple timeseries simultaneously by looking at the combination of values at each time step and if their specific combination is outside of the normal range.

Google BigQuery offers other anomaly detection models for 'independent and identically distributed random variables'. These are variables with the same probability distributions which are mutually independent. The monitored water quality variables are not independent, for example, temperature affects DO and pH affects ammonium. However, the decision was taken to apply these models in the benchmarking, as they work in the same way as the Isolation Forest in the RDMAI model - they highlight an anomaly in the combination of values in a row, i.e., anomalies over multiple water quality parameters within a single timestep. In this way, these models are true multivariate anomaly detection models. However, these models do not remove trend and seasonality of the time series before performing anomaly detection and do not treat the data as a time series. The models applied are:

1. https://docs.cloud.google.com/bigquery/docs/reference/standard-sql/bigqueryml-syntax-create-kmeans

2. https://docs.cloud.google.com/bigquery/docs/reference/standard-sql/bigqueryml-syntax-create-pca

- K-means: 'Detect anomalies based on the shortest distance among the normalized distances from the input data to each cluster centroid' [1].

- PCA: 'Detect anomalies based upon the reconstruction loss in terms of mean squared error'[2].

When running ML.DETECT_ANOMALIES for the ARIMA_PLUS_XREG model, you can specify the parameter 'anomaly_prob_threshold' as a fraction (0 to 1), which specifies the probability threshold for identifying anomalies in your data.

In this benchmarking exercise, dissolved oxygen (DO) and ammonia were chosen as the main variables to forecast using ARIMA_PLUS_XREG, with all other timeseries (pH, turbidity, conductivity and temperature) used as external regressors. The choice of DO and ammonia as the main timeseries is based on these two variables being dependent on the other parameters, whereas pH, turbidity, conductivity and temperature are relatively independent variables. After some experimenting, an anomaly_prob_threshold of 0.95 was chosen. Lower values of anomaly_prob_threshold, such as 0.6, resulted in the ML.DETECT_ANOMALIES detecting a lot of anomalies in the analysed data. All monitoring data provided for the three sites were used to construct the ARIMA_PLUS_XREG models (see Table 1.1).

For the PCA models, you need to specify the number of principle components ('NUM_PRINCIPAL_COMPONENTS'). This was set to 2 because most variability in the data is usually explained in the first 2 principal components.

For the K-means models, you need to specify the number of clusters ('NUM_CLUSTERS'). This refers to the number of clusters to identify in the input data, which relates to pollution events. This was set to 8 since the number of pollution events are not known beforehand.

In summary, the anomaly detection methods used in Google BigQuery were:

- ARIMA_PLUS_XREG

- K-means

- PCA

Microsoft Azure Data Explorer

Description

The Data Explorer is a Microsoft Azure data analytics service. It can be used to analyse, in real time or near real time, large volumes of data. These data can be streamed from applications, websites or Internet of Things (IoT) devices, such as sondes. Manipulating and querying data on the Data Explorer uses an SQL-type language. The Data Explorer operates in the cloud, meaning that there is no need to install or manage.

Anomaly detection is offered among the suite of data analysis tools in the Data Explorer. These include methods for univariate and multivariate anomaly detection. Since we are interested in multivariate anomaly detection in this benchmarking report, the univariate anomaly detection methods offered in the Data Explorer are not described further. The multivariate anomaly detection methods offered in the Data Explorer include:

series_mv_ee_anomalies_fl(): Uses an elliptic envelope model to detect anomalies by considering multiple metrics simultaneously.

series_mv_oc_anomalies_fl(): Detects anomalies using the One Class Support Vector Machine (SVM) model.

series_mv_if_anomalies_fl(): Detects anomalies using Isolation Forest model

Costing

Using an on-demand pricing model, the costs for using Azure Data Explorer are based on data ingestion volume, storage duration, and compute resources used. There is a website (https://azure.microsoft.com/en-us/pricing/calculator/?service=data-explorer) for estimating costs. The Data Explorer can be tried for free for a year.

Anomaly detection used

Unfortunately, while univariate anomaly detection is offered in the free Data Explorer trial, the multi-variate models are only available in the paid-for version. Therefore, this benchmarking exercise did not evaluate the Azure Data Explorer option further.

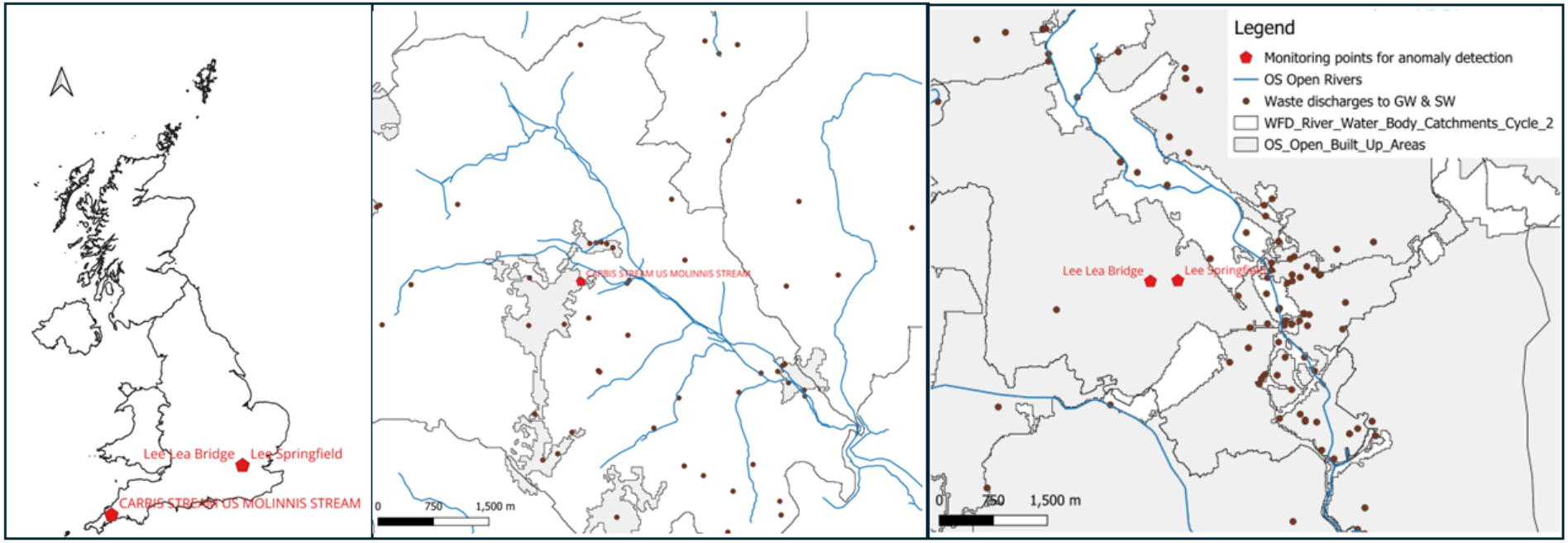
1.2 Monitoring points for anomaly detection

Data used for anomaly detection are from three high-frequency Environment Agency sondes, as part of the network of high-frequency monitoring points viewable in the Hydrology Data Explorer (https://environment.data.gov.uk/hydrology/explore). Table 1.1 and Figure 1.1 provide a summary of the monitoring point data used. The benchmarking was constrained to these relatively short samples of monitoring data from a limited number of monitoring points because they include sections of monitoring data that were manually labelled. The manual labelling provides some ‘correct’ anomaly detection to benchmark the models against.

Table 1.1 Summary of the EA high-frequency water quality monitoring points used

Name	EA Reference	Lat	Long	Date Range
Lee Springfield	E01336A	51.54975	-0.03708	June 2019 to April 2025
Lee Lea Bridge	E26020A	51.54975	-0.04257	Oct 2020 to April 2025
CARBIS STREAM US MOLINNIS STREAM	E12040A	50.40078	-4.78728	Nov 2017 to Feb 2020

Figure 1.1 Position of monitoring points selected for anomaly detection



1.3 Benchmarking Methodology

The benchmarking method used in this report is an extension of the RDMAI Anomaly Detection Model validation. The benchmarking uses the same water quality monitoring data for three sites used in the model validation, along with manual labelling by domain experts to serve as a reference for evaluating model performance. The benchmarking in addition evaluates the industry-standard model in the same way: anomaly detection in Google BigQuery. The assessment is limited to anomaly detection of each model, and not to the filtering of anomalies via clustering that the RDMAI model offers (i.e., up to '*anomaly detection*' in 1.1.1). This is because the information provided by the clustering in the RDMAI model at this point remains limited, as clusters are not yet linked to a type of event.

Within the RDMAI model, the anomaly detection is run for a year's worth of data for each site (the minimum data length recommended by the model documentation), up to the *anomaly detection* stage described in section 1.1.1, i.e., without further clustering or filtering of detected anomalies.

For the Google BigQuery anomaly detection:

- The monitoring data are averaged to hourly: this is because hourly averaging of data is part of the RDMAI workflow, and comparison of anomalies between the two models requires a consistent dataset.

- The construction of the ARIMA_PLUS_XREG, K-Means and PCA models uses all available monitoring data.

- Among the monitored parameters, DO and ammonia would more likely be dependent parameters, while pH, temperature and turbidity would be independent. Therefore, DO and ammonia were chosen the main timeseries to predict in two separate ARIMA_PLUS_XREG models. Then, for a dataset and timestamp, if the ARIMA_PLUS_XREG models identified either DO or ammonia as anomalous, the timestamp was flagged as anomalous.

- When running ML.DETECT_ANOMALIES for the ARIMA_PLUS_XREG model, an anomaly_prob_threshold of 0.95 is used.

- In the construction of the PCA models, 'NUM_PRINCIPAL_COMPONENTS' is set to 2.

- In the construction of the K-means models, 'NUM_CLUSTERS' is set to 8.

- When running ML.DETECT_ANOMALIES for the K-means and PCA models, the 'contamination' parameter is set to 0.1 – this is the proportion of anomalies in the training dataset.

For the monitoring data labelled by domain experts, manually labelled datasets falling within the time window of model analysed data were identified. In other words, it was ensured that

manually labelled data corresponded temporally the data analysed by the different models. In the labelled datasets, the domain experts could flag each parameter at each reading individually as an anomaly. This benchmarking flagged the entire reading (corresponding to a particular timestamp) of manually labelled data as anomalous if any of the individual parameters were marked as anomalous. This was done because the RDMAI model anomaly detection is multivariate.

The labelled anomalies and model-detected anomalies were re-sampled to every three hours and compared. This was done because of inconsistencies in which domain experts and models labelled anomalies: anomalies could be labelled at the start, middle or end of a pollution event, or a combination of these. In the resampling, if any anomaly was detected over the three-hour period, the entire three-hour period was flagged as anomalous. The choice of three hours in which to resample the data is reasonable given the durations of events in the labelled datasets. During the comparison, model true positives, true negatives, false positives, and false negatives were computed, by comparison to the labelled data.

1.4 Benchmarking Limitations

This model benchmarking only considered three monitoring datasets. The available datasets for benchmarking were limited by those that were manually labelled by domain experts. A more comprehensive benchmarking could consider further datasets that are more representative of the diverse geographical and climatological conditions across the UK.

This benchmarking identified a single platform, Google BigQuery, and three industry-standard models that could be compared to the RDMAI model. Barriers to using further models for benchmarking included cost and time. Other anomaly detection models are known to exist, but it was not possible to benchmark with these due to time, resource, and access constraints. Some of these models are mentioned in Section 1.1.2.

The ARIMA_PLUS_XREG anomaly detection in Google BigQuery is different to the RDMAI model in that it is used for forecasting, with anomaly detection being prediction based. Therefore, the model requires a main timeseries to be set for prediction and anomaly detection, which in this case was chosen based on domain knowledge of relationships between water quality variables.

The K-means and PCA anomaly detection models in Google BigQuery are likely not entirely suitable for application to the water quality monitoring datasets as the monitored variables are not independent. In addition, setting the parameter ‘NUM_CLUSTERS’ for the K-means model is uncertain as this can be related to the number of pollution events in a dataset and this is not known beforehand. Also, the ‘contamination’ parameter in ML.Detect_Anomalies was set to 0.1, assuming 10% of data in the datasets was anomalous. This was however an assumption as we don’t have manual labelling across the entire datasets.

Within the data manually labelled by domain experts, approaches to labelling data were different among the different domain experts. This resulted in inconsistencies in detected anomalies among the different labelled datasets. In addition, domain experts identified

anomalies within individual parameters – making their anomaly detection univariate which is consistent with the anomaly detection performed by Google BigQuery's ARIMA_PLUS_XREG. In contrast, the algorithm created for RDMAI as well as the Google BigQuery K-means and PCA models are forms of multivariate anomaly detection. This difference between outputs was overcome by assuming an entire reading (timestamp) within the labelled data to be anomalous if any of the individual parameters were flagged as anomalous by the domain experts.

While ARIMA_PLUS_XREG is a multivariate forecasting model, it does not identify anomalies across multiple timeseries simultaneously, as the RDMAI model does. It just uses other variables to aid in the prediction of the main target variable. Therefore, this benchmarking identified anomalies in ARIMA_PLUS_XREG using DO and ammonia as the main timeseries in two separate models, after which the model detected anomalies for the two parameters were combined.

The approach taken to quantitatively compare model performance is not entirely appropriate and as does not accurately reflect model performance because the manual labelling is pattern-based rather than point-based. Therefore, the results of the performance evaluation should not be regarded as absolute indicators of model performance, but rather of relative model performances.

2. Benchmarking Results

2.1 Model 'Ease-of-Use'

Table 2.1 Comparison of model 'ease-of-use'

Model Outputs	Google BigQuery (Existing Industry Model)	RDMAI Model
Installation & Run Process	There is no installation process for Google BigQuery. Once the user has an account, the Data Explorer can be accessed from a Web browser. Users need a basic knowledge of SQL to create tables, import data and run queries. Google provides comprehensive documentation and tutorials. Running anomaly detection in BigQuery similarly takes seconds, depending on the size of the dataset.	Installing and running the RDMAI Anomaly Detection Model requires basic knowledge of Python. Once installed, running the model is easily done using the provided Jupyter Notebook. Some data processing of the input water quality monitoring data might be required, such as ensuring the date column is in the right format, and that the columns have valid headers.
Data inputs	Any multivariate timeseries data.	Time series high-frequency water quality monitoring data for pH, dissolved oxygen (mg/L or %), turbidity (NTU), temperature (°C), and ammonia (mg/L). Conductivity can also be included.
Licensing Requirements	Google BigQuery is part of their cloud services offering, and so is a paid-for service. The cost is dependent on use.	The RDMAI Anomaly Detection model is free and open.

2.2 Model Outputs

Table 2.2 Comparison of model outputs

Model Outputs	Google BigQuery (Existing Industry Model)	RDMAI Model
Data Format	A timeseries of Boolean flags, indicating an anomaly or no anomaly at each timestep.	A timeseries of Boolean flags, indicating an anomaly or no anomaly at each timestep. Additionally, the scripts provide some supporting graphs, such as the seasonal and trend signals in a timeseries, and anomalies in relation to the dataset residuals.
Temporal Resolution	The timeseries of anomalies are provided at the temporal resolution of the input timeseries data.	The RDMAI model takes in high-frequency water quality monitoring data and averages these data to an hourly frequency. Therefore, outputs of anomalies are at an hourly frequency.
Geographic Constraints	The anomaly detection in Google BigQuery has no geographic constraint.	The anomaly detection in the RDMAI model has no geographic constraint.

2.3 Time and Cost Comparison

Google BigQuery

Time

BigQuery uses SQL for running queries, which presents a learning curve barrier to novices. However, once familiar with the platform, making tables, bringing in data, and running the anomaly detection queries take seconds.

Cost

BigQuery is part of Google's Cloud Services offering, which is a paid-for service. On the 'on-demand' payment plan, costs depend on usage and can quickly escalate. On-demand queries cost $6.25 per TB processed. It's difficult to come up with a rigorous estimate of cost to a water company for data processing (ignoring telemetry/transmission costs, staff time, etc.). Assuming a water company has 4,000 sondes, anomaly detection every hour based on the previous years' worth of high-frequency data (assuming a data volume of 700 kB), would equate to approximately 9,000 TB of data processed per year. At $6.25 per TB processed, this would equate to a yearly cost of around $56,250 or £42,073. However, the anomaly detection could probably be optimised to look at less data at a time.

There are also costs for storage, but these are not applicable as storage would possibly be needed regardless of the model applied.

RDMAI Anomaly Detection Model

Time

Installing and running the RDMAI Anomaly Detection Model requires a basic knowledge of Python and will take between 1 and 2 hours. Once the model is installed, running the anomaly detection model on around 2 months of data takes seconds.

Cost

The model is free and open, so there are no costs, besides for costs associated with data storage if the model is integrated within a monitoring system.

2.4 Model Performance

Please note that it is difficult to obtain a definite quantitative accuracy assessment of the models. The confusion matrix shown in Table 2.3 is not entirely appropriate and would not accurately reflect model performance because the manual labelling is pattern-based rather than point-based. Therefore, the statistics shown in Table 2.3 should not be regarded as absolute indicators of model performance, but rather of relative model performances.

Figure 2.1 shows the labelled anomalies and those identified in Google BigQuery ARIMA_PLUS_XREG, PCA and K-means models and in the RDMAI model for Lee Lea Bridge.

As shown in Table 2.3, The RDMAI and Google BigQuery ARIMA_PLUS_XREG and K-means models all identified similar percentages of anomalous events (when compared to the labelled data) ranging between 23.7% and 26%, whereas the Google BigQuery PCA model identified less (14.2%)

Figure 2.2 shows the labelled anomalies and those identified by the RDMAI model and Google BigQuery ARIMA_PLUS_XREG, K-means and PCA models for Lee Springfield. There is a general correlation between clusters of labelled anomalies and RDMAI and ARIMA_PLUS_XREG identified anomalies. The PCA model identified many more anomalies, whereas K-means identified far fewer anomalies. Table 2.3 shows that the PCA model identifying the highest proportion of anomalous events (49%), followed by the ARIMA_PLUS_XREG model at 29.8% and the RDMAI model at 20.2%, and the K-means model performing least well at 9.1%

Figure 2.3 shows labelled data and model identified anomalies for Carbis Stream us Molinnis Stream. The manual labelling identified many anomalies for this monitoring point during the period assessed. These higher number of anomalies are reflected in the RDMAI, ARIMA_PLUS_XREG, and PCA identified anomalies, whereas K-means identified fewer anomalies. Table 2.3 shows the RDMAI and ARIMA_PLUS_XREG identified the most anomalies, at 60.4% and 62.3%, respectively, followed by the PCA model at 47.9%, with the K-means model identifying the least anomalies at 23.1%

Figure 2.1 Comparison between model multivariate anomaly detection and labelled data at Lee Lea Bridge

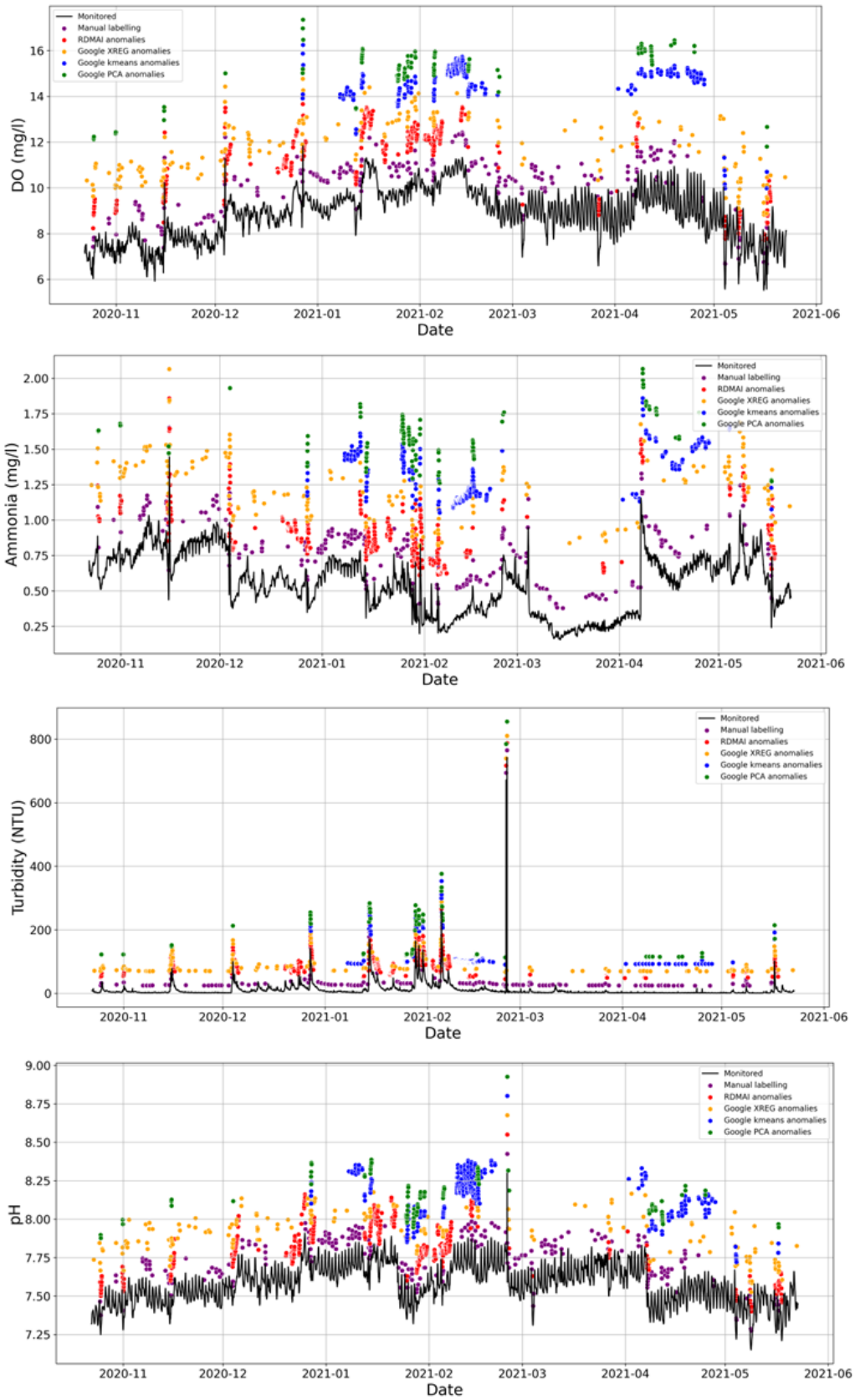


Figure 2.2 Comparison between model multivariate anomaly detection and labelled data at Lee Springfield

Figure 2.3 Comparison between model multivariate anomaly detection and labelled data at Carbis Stream us Molinnis Stream

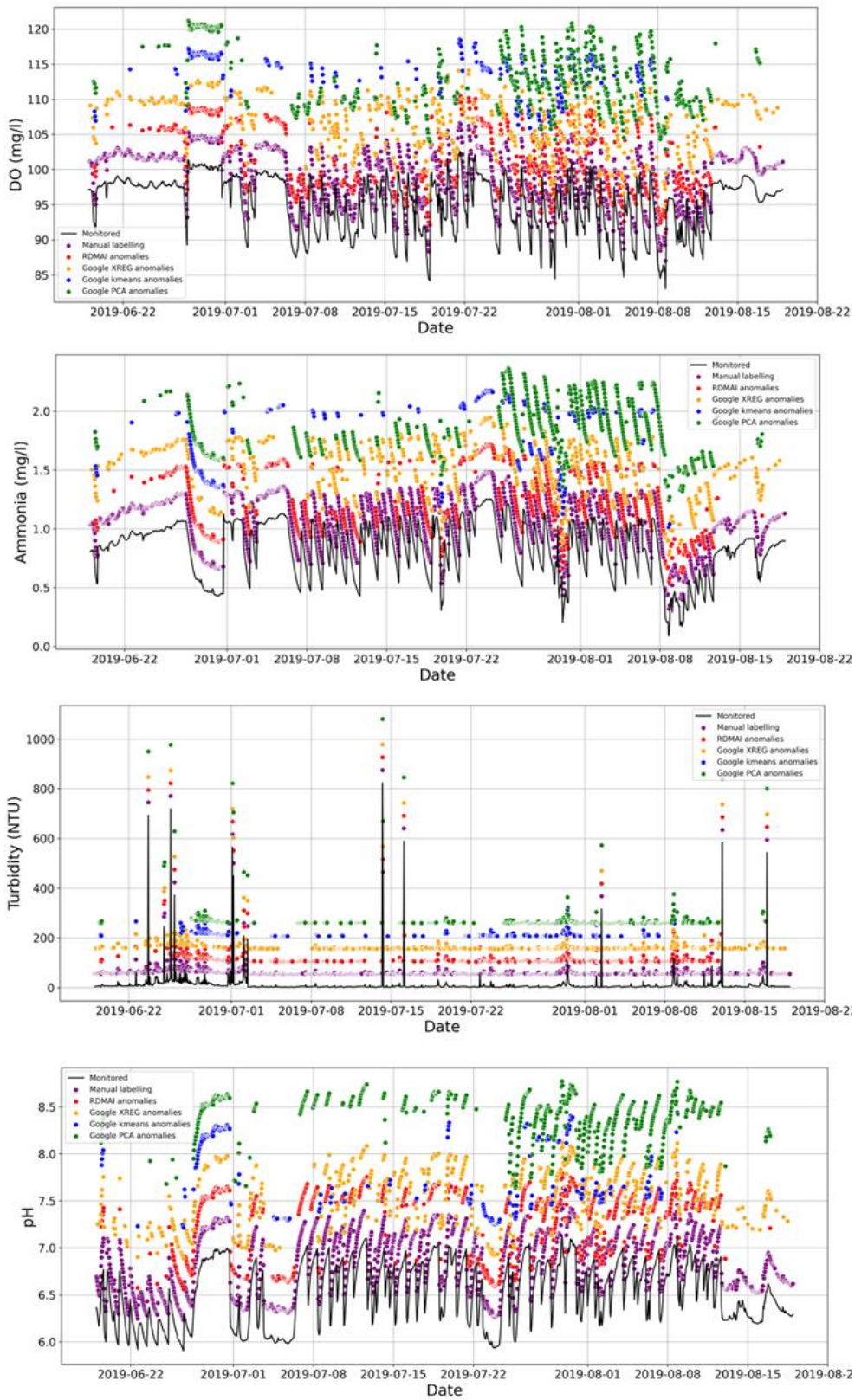


Table 2.3 Summary of model performances when compared to labelled data. Please node, a 3-hour buffer window was applied to both manually labelled and modelled anomalies to account for inconsistencies in where in pollution events anomalies are flagged

Model	Site	Date range	n	False positives	False negatives	True positives	True negatives	Percentage anomalous events identified
RDMAI Anomaly Detector	Lee Lea Bridge	Oct 2020 June 2021	1701	165	125	44	1367	26.0%
Google ARIMA_PLUS_XREG	Lee Lea Bridge	Oct 2020 June 2021	1701	106	126	43	1426	25.4%
Google K-means	Lee Lea Bridge	Oct 2020 June 2021	1701	111	129	40	1421	23.7%
Google PCA	Lee Lea Bridge	Oct 2020 June 2021	1701	38	145	24	1494	14.2%
RDMAI Anomaly Detector	Lee Springfield	June 2019 Aug 2019	477	10	166	42	259	20.2%
Google ARIMA_PLUS_XREG	Lee Springfield	June 2019 Aug 2019	477	38	146	62	231	29.8%
Google K-means	Lee Springfield	June 2019 Aug 2019	477	30	189	19	239	9.1%

Google PCA	Lee Springfield	June 2019 Aug 2019	477	125	106	102	144	49.0%
RDMAI Anomaly Detector	Carbis Stream us Molinnis Stream	June 2019 Aug 2019	488	6	190	290	2	60.4%
Google ARIMA_PLUS_XREG	Carbis Stream us Molinnis Stream	June 2019 Aug 2019	488	4	181	299	4	62.3%
Google K-means	Carbis Stream us Molinnis Stream	June 2019 Aug 2019	488	0	369	111	8	23.1%
Google PCA	Carbis Stream us Molinnis Stream	June 2019 Aug 2019	488	7	250	230	1	47.9%

3. Benchmarking Conclusion

3.1 Summary

A summary table of key model characteristics can be found at the start of the report.

3.2 Conclusion

This benchmarking faced various limitations and challenges, including:

- Limited datasets for analysis.
- Due to time and access constraints, only three industry-standard models were assessed on a single platform (Google BigQuery).
- Inconsistencies in labelling between the different labelled datasets.

There are fundamental differences in the approaches in which the compared models identify anomalies, which is a strength of this benchmarking, as it highlights advantages and disadvantages of the different approaches.

The clusters of anomalies identified by the RDMAI model broadly lined up with those by Google BigQuery ARIMA_PLUS_XREG and PCA, and with labelled data. The labelled data showed a larger number of isolated anomalies than the RDMAI model, which the Google BigQuery ARIMA_PLUS_XREG represented better, at least in terms of quantity.

Table 2.3 shows a comparison of the model-identified anomalies with labelled anomalies when the datasets were resampled to every 3 hours. Please note that because of the difficulties in comparing the pattern-based manual anomaly labelling with point-based modelled anomalies, the statistics shown in Table 2.3 should not be regarded as absolute indicators of model performance, but rather of relative model performances.

Table 2.3 shows that the RDMAI and Google ARIMA_PLUS_XREG and PCA models have a similar overall performance and identified relatively small proportions of labelled anomalous events (RDMAI: 20.2% to 60.4%; ARIMA_PLUS_XREG; 25.4% to 62.3%; PCA: 14.2% to 49%). However, the Google K-means model identified the least labelled anomalous events (9.1% to 23.7%). The observations for the RDMAI model are consistent with that of the validation, showing that the RDMAI model struggles to identify low-magnitude events as anomalies, but can generally detect high magnitude deviations from a baseline. However, overall, the RDMAI model showed a similar performance to the Google models for the datasets assessed.

Although the RDMAI model is implemented as a stand-alone anomaly detection pipeline, this does not prevent it from being deployed on cloud infrastructure or used for near real-time processing. With appropriate engineering—such as containerisation and integration into cloud

compute services—the model can operate continuously or on streaming data. In contrast to fully managed commercial services, RDMAI offers greater flexibility in configuration, full transparency into each processing stage, and improved interpretability of the underlying seasonality, trends, and anomaly-detection logic. Thus, while commercial providers offer convenience and out-of-the-box scalability, the RDMAI model is not technically limited; instead, it provides deeper control and explainability when engineered and deployed correctly.

The RDMAI anomaly detection model offers various advantages over the assessed industry standard model, including:

- A specific focus on water quality timeseries data.
- A multivariate approach to identifying anomalies that is different to the forecast, predictive approach of ARIMA_PLUS_XREG and arguably better suited for application to high frequency water quality monitoring data. This is because the focus of focus of Section 82 monitoring is not forecasting but reacting timeously to pollution events. In addition, as a forecasting model, the anomaly detection in ARIMA_PLUS_XREG is in a single specified main timeseries, whereas the RDMAI model identifies anomalies in multiple timeseries simultaneously.
- The RDMAI model is free and open, and so can be easily adapted for incorporation by water companies and regulators into their systems managing high-frequency monitoring data.

In addition, the RDMAI anomaly detection model overcomes several challenges associated with traditional approaches by providing a framework that enables more precise estimation of long-term trends and recurring seasonal patterns.

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